Sales Data Pipeline: SQL Cleaning & Python Analytics

Summary Map : Raw Data ➔ Python ➔ SQL (Clean) ➔ CSV ➔ Python (Predict)

Step 1: Ingesting & Loading (Python to PostgreSQL)

In this phase, Python served as the primary engine for handling raw data. A script was developed to ingest the raw CSV file and load it directly into a PostgreSQL database. Instead of a simple file import, a persistent connection was established that "poured" the data into a structured table. An automated "bridge" was created using the SQLAlchemy library to facilitate the data transfer.

Step 2: Transformation & Cleaning (The SQL Workspace)

Once the data was stored inside PostgreSQL, SQL was used for deep cleaning and organization. The process went beyond simple row deletion; the entire dataset was transformed. This stage involved fixing missing values, standardizing text formats, and ensuring all data types were accurate—such as converting text-based dates into proper Date formats.

Step 3: Extraction & Analytics (SQL to Python)

With the data fully cleaned, it was extracted from the database. A final version was saved as a CSV file to keep a permanent record of the "clean" data. This file was then imported back into Python to begin the analytical phase.

Step 4: Machine Learning & Predictive Modeling

Once the cleaned dataset was prepared, the project transitioned into the predictive modeling phase. Two distinct algorithms were implemented and compared: **Linear Regression** (as a baseline) and a **Random Forest Regressor**. To ensure categorical data was compatible with these models, **One-Hot Encoding** with drop_first=True was applied, effectively preventing the "Dummy Variable Trap" and maintaining mathematical stability. This stage highlighted that sales patterns are **non-linear**, as the ensemble-based approach of the Random Forest was better suited to capture the complex interactions between price, quantity, and product categories.

Step 5: Evaluation & Performance Insights

The final phase involved a rigorous evaluation using the **R^2 Score** as the primary performance metric. By calculating the **percentage of variance** explained by each model, the **Random Forest** was identified as the superior performer, achieving an **R^2 of 0.44**. A visualization of "Actual vs. Predicted" sales values confirmed that the model accurately tracks core trends. However, the presence of "outliers"—data points located far from the identity line—indicates that external variables not present in the current dataset, such as seasonality or specific marketing promotions, continue to influence high-value sales results.